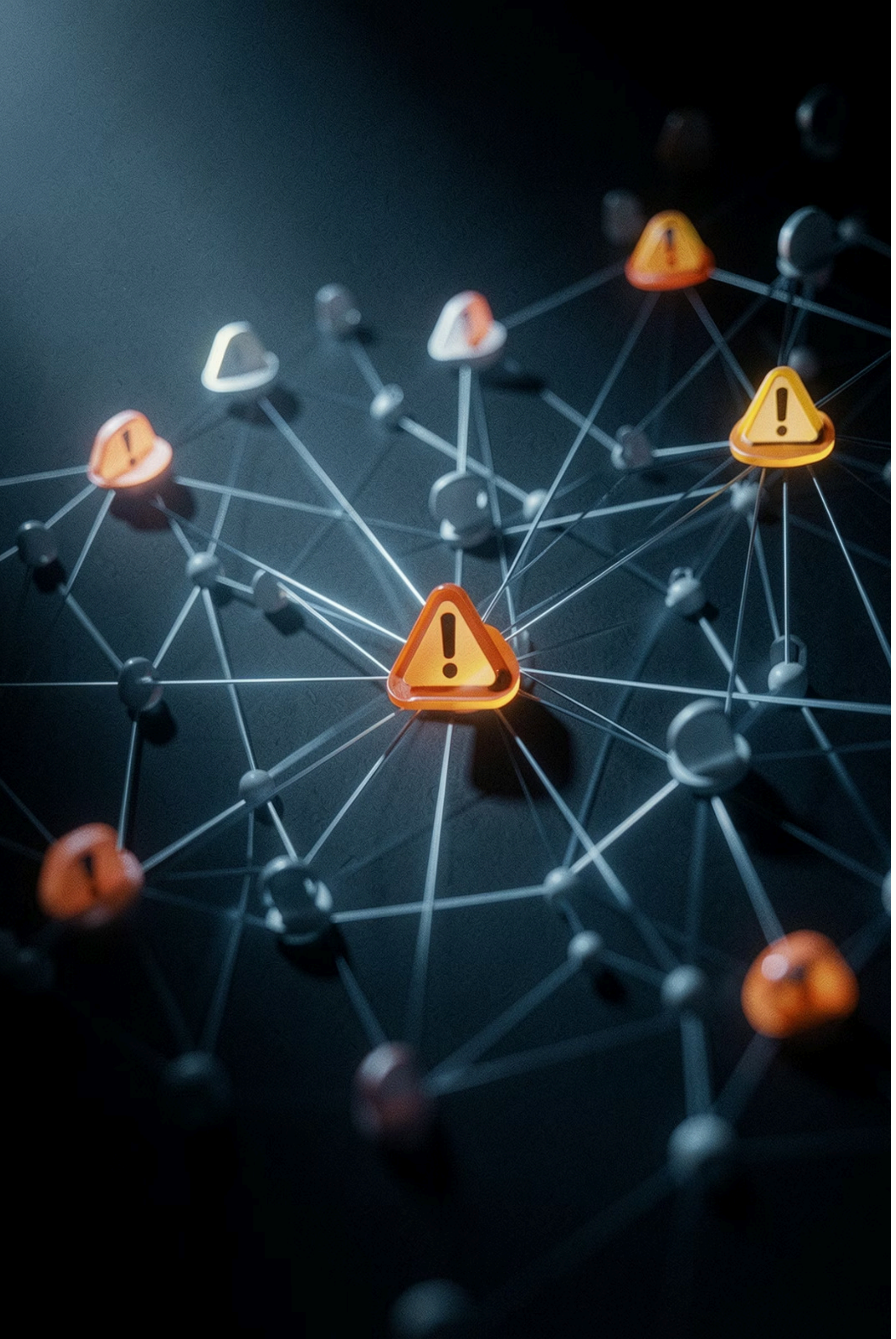
Banking Fraud Detection

Machine Learning Analysis Report — **Vaheeson Vasanthakumar**

RANDOM FOREST CLASSIFIER | 284,807 TRANSACTIONS | 2023 DATASET



The Fraud Challenge

Scale

Billions of transactions annually — manual review impossible

Complexity

Fraudsters use advanced, evolving techniques to bypass security

Solution

ML models learn from historical data and adapt to new patterns

Project Objectives

01

EDA

Explore transaction data distribution

02

Preprocess

Clean data, handle class imbalance

03

Model

Build high-accuracy fraud classifier

04

Evaluate

Precision, recall, F1, confusion matrix

05

Deploy

Actionable production recommendations

Dataset at a Glance

284,807

Total transactions (Sept 2023)

31 Features

30 PCA-transformed + 1 target

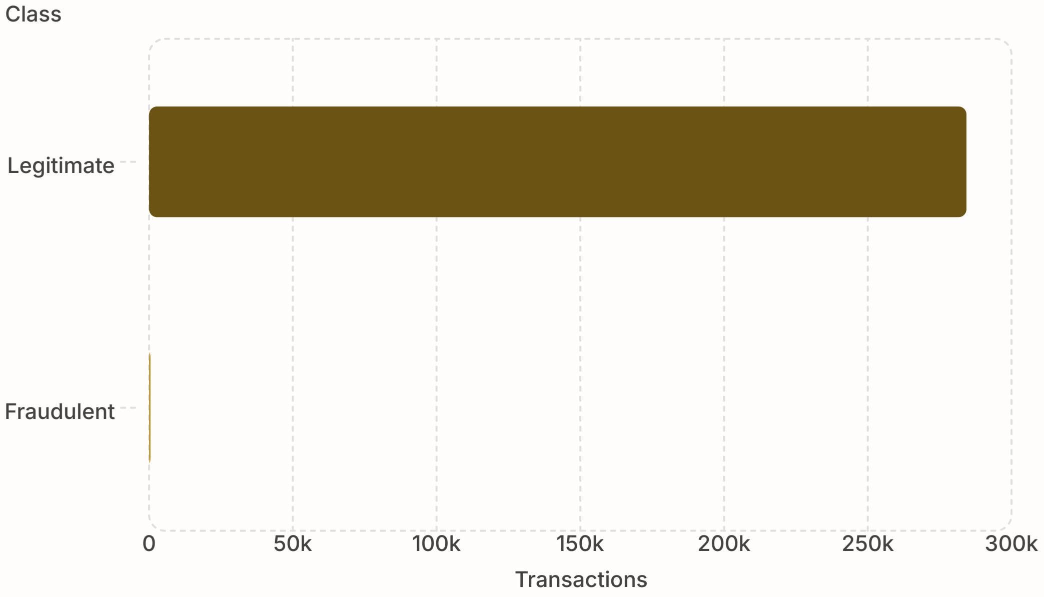
99.827%

Legitimate transactions

0.173%

Fraudulent transactions (492 cases)

The Class Imbalance Problem



Why It Matters

- Models can achieve high accuracy by predicting all transactions as legitimate
- Fraud patterns may go undetected
- Standard accuracy becomes misleading

Solution: Stratified train-test splitting preserves class ratio in both sets

Methodology

1. Preprocess Data
2. Exploratory Analysis
3. Stratified Split
4. Random Forest

Stratified splitting ensures the 0.173% fraud rate is preserved across training and test sets, enabling the model to learn from representative fraud samples.

Why Random Forest?

Robustness

Handles outliers and diverse fraud patterns

Feature Importance

Built-in ranking of most predictive features

Ensemble Power

100 trees reduce overfitting, improve generalization

Efficiency

Processes 30 features and 227K+ samples efficiently

Model Performance

99%

Overall Accuracy

99 of 100 transactions correctly classified

89%

Fraud Precision

89 of 100 flagged transactions are truly fraud

85%

Fraud Recall

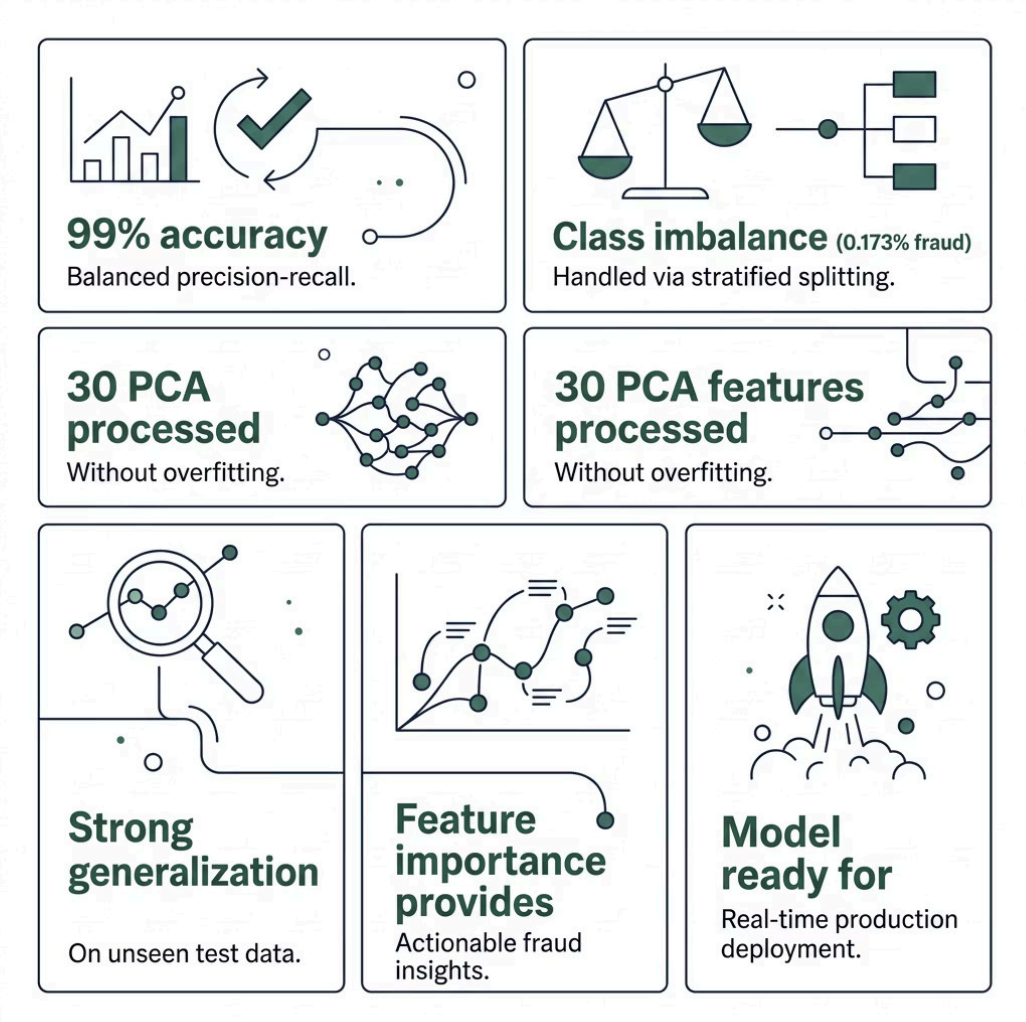
85 of 100 fraud cases successfully caught

0.87

F1-Score

Balanced precision-recall for fraud class

Key Findings



Bottom Line

The Random Forest classifier achieves exceptional fraud detection performance despite severe class imbalance — a practical, production-ready solution for banking environments.

Deployment & Next Steps

1

Deploy Now

Real-time monitoring with automated fraud alerts

2

Monitor

Track precision, recall, F1 on live transactions

3

Retrain

Monthly/quarterly updates to adapt to new fraud patterns

4

Optimize

Hyperparameter tuning, feature engineering, ensemble methods